Jacob Epshteyn

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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA
S.B. in Electrical Engineering with Computing Expected May 2029
GPA: 4.5/5.0 | **Relevant Coursework:** Introduction to Algorithms, Linear Algebra, Low-Level Programming

Technical Projects

Auralis – Interactive Electronic Instrument 2023–2025

- Designed and built a touchless musical instrument integrating 12 time-of-flight distance sensors, embedded controllers, audio synthesis, and programmable lighting in a custom enclosure.
- Developed a dedicated Raspberry Pi Pico sensing subsystem with dual I^2C buses and serial communication to a Raspberry Pi 4 for responsive hand-controlled interaction.
- Implemented configurable note and chord triggering, instrument playback, a graphical control interface, and synchronized lighting effects across 156 addressable LEDs.
- Modeled and fabricated a modular illuminated frame, iterating on sensor placement, wiring, and assembly; demonstrated the completed instrument in a live graduation performance.

34-String Lever Harp – CAD and Fabrication MIT Hobby Shop, 2025–2026

- Reconstructed a five-foot lever harp in Fusion 360 from dimensioned plans and scaled reference drawings; modeled individual components and verified their fit in a complete digital assembly.
- Fabricated the neck, pillar, soundboard, body, and other components using CNC machining and traditional woodworking; prepared stock and precision-located string and hardware holes.
- Diagnosed repeated failure of the soundboard adhesive joint under string tension; after an unsuccessful re-gluing attempt, reinforced the assembly with concealed screws, preventing further separation.

Experience & Leadership

Boston Urban Art Association Boston, MA
Co-Creator, Writer, and Workshop Facilitator *Jun 2024–Aug 2025*

- Co-created an independent Boston street-art publication; co-wrote 10,000+ words of research-based visual analysis and local arts commentary.
- Presented BUAA's vision for mapping Boston street art to Bay Cove Academy students and facilitated an urban-art workshop using 3D-printed buildings and paint pens.

Game Development and Club Leadership 2017–2025
Independent Roblox Developer; High-School Club Founder

- Founded and led a game development club, teaching Roblox Studio, programming fundamentals, and collaborative development; independently developed Roblox game systems and interfaces across projects reaching millions of plays.

Boston Bikes Boston, MA
Youth Instructor *Jun 2024–Aug 2024*

- Taught cycling safety to 25+ children ages 8–12, supervising group rides and supporting program logistics.

Technical Skills

Programming:	Python, JavaScript, Lua/Luau, HTML/CSS
Embedded/Electronics:	Raspberry Pi, Raspberry Pi Pico, Arduino, serial and I^2C communication, soldering, electronics prototyping
CAD/Fabrication:	Fusion 360, Onshape, CNC machining, 3D printing, laser cutting, woodworking
Other Tools:	Blender, Figma, Adobe Creative Suite

Honors & Awards

- Winner, Desmos International Math Art Competition (2022); 1st Place, Northeastern University Calculus Field Day – Precalculus (2024); 1st Place, Harvard Academies Web Design Competition (2023).